Springer Finance

2026 FINTECHATHON · INTERNATIONAL TRACK · TOPIC E — SME FINANCE COPILOT

The finance copilot for small businesses that run on cash and mobile money

Every department of the business in one app, offline-first, with an AI agent that analyses the books, forecasts cash, matches financing, and prepares records a person approves.

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THE PROBLEM

A small business owner in Yaoundé runs three activities and sees none of them clearly

1

No daily picture

Sales sit in cash drawers, notebooks and mobile-money histories. The owner learns about a loss weeks later, if at all.

2

Cash leaks

Nobody can say how much cash each manager should be holding tonight, or what customers still owe on credit.

3

No records, no credit

Microfinance and banks ask for months of records and statements. Without them, the owner turns to tontines and expensive nano-loans.

Generic accounting software assumes an accountant, a bank account, a stable connection and one activity. This owner has none of the four.

THE MARKET

Cameroon is a country of small businesses and mobile money

443 524

SMEs in 2024, **99 %** of all enterprises; 97 % are individual businesses

30.9 M

mobile-money accounts in Cameroon (65 % of the CEMAC region)

26 773 bn

FCFA moved by mobile money in Cameroon in 2024

Beachhead

Owners with several activities (shop, bar, guest house, venue, services) in the big cities, led by Douala and Yaoundé, which hold 57 % of SMEs.

Then

The CEMAC and wider francophone Africa share the FCFA and the OHADA accounting framework (SYSCOHADA) that the app already implements.

Sources: Ministry of SMEs 2024 statistical yearbook (via Business in Cameroon, May 2025); BEAC payment services report 2024 (via Ecofin Agency); IFC: microenterprises in Cameroon face a financing gap larger than most of West and Central Africa.

THE SOLUTION

One app for the whole business, and a copilot that understands it

B

For the owner

Live view of every department: sales, expenses, profit, cash each drawer should hold, debts. Daily reports to approve, cash handovers to confirm, statements, optional SYSCOHADA accounting with VAT.

D

For each department

13 department types, each with its own screens: till (shop, bar, restaurant), bookings (venue, guest house), tickets (pressing, car wash), batches (bakery, farm), memberships (gym). It works offline and syncs later.

AI

Finance Copilot

Ask in English or French. It reads the business's own records through tools, forecasts cash, raises alerts, ranks financing options, and prepares records the person approves.

✓ Offline-first

✓ English and French

✓ FCFA, mobile money, credit sales

✓ SYSCOHADA and VAT when ready

MEET THE COPILOT · REAL OUTPUT FROM THE SANDBOX RUN

“Will we run short of cash in the next 30 days?”

Mme Ngono · owner

Will we run short of cash in the next 30 days? Besides the drawers we have 350 000 FCFA on Orange Money.

`forecast_cash_flow · READ`

Finance Copilot

No, you will not run short of cash in the next 30 days.

- **Lowest projected balance: 562,223 FCFA** (today: 350,000 on Orange Money + 212,223 in department drawers).
- **Expected balance in 30 days: 2,102,311 FCFA** (or 1,331,705 FCFA under the cautious scenario).
- **First negative day:** none.

Every figure is computed, not generated

The model chooses the tool and explains the result. The forecast itself is a tested function: the last 12 weeks, by weekday, with an expected and a cautious scenario (−15 % in, +10 % out).

Checked automatically

In the sandbox run, **69 of 71** figures in the answers were found literally in the tool data. The other two were a date and a correct multiplication.

TOPIC E, POINT BY POINT

Financing matching, cash-flow alerts, business analysis

Financing options

“I need 1 500 000 FCFA to restock the bar, over 6 months.”

Microfinance loan: **267 789 FCFA/month**, within 40 % of average net cash · njangi / tontine: 250 000/month, no interest · supplier credit 30–60 days · collect first the **462 450 FCFA** customers owe (31 % of the need).
Indicative; not an offer or advice.

Alerts

“Give me today's alerts.”

High: loss of 115 798 FCFA this month.
High: 393 050 FCFA owed for over 30 days, 208 000 over 90.
Medium: 4 handovers (569 000 FCFA) waiting 2+ days for confirmation.
Also: cash running out, costs rising faster than income.

Analysis

“Which department is weaker, and why?”

Bar Le Carrefour: −152 875 FCFA this month (in 460 650, out 613 525); costs rose faster than sales. The shop made +37 077 FCFA. Next steps: review bar overheads; compare stock used with sales.

Real answers from the sandbox run. Daily, weekly and monthly reports and morning alerts also reach the owner by e-mail.

IT ACTS, BUT NEVER ALONE

The agent prepares; the person approves; the app records



READ · 8 tools	Figures of the departments **this person** may see. Nothing else.
PROPOSE · 3 tools	**Writes nothing.** It validates the request and stores a proposal whose summary is written by server code. Approval runs it once, as the person, through the app's normal write path.
NO TOOL	Void, delete, settings, people, passwords, confirming cash. The agent cannot do them, whatever it is told.

SECURITY AND COMPLIANCE

Built for money from day one

6 / 6
attacks on the agent held: deleting sales, other departments' data, a planted prompt injection, cash above the drawer...

687
attempts to touch another business's or department's records, all refused

100 %
of changes through one validated, idempotent, audited write path

0
ways to edit a posted ledger entry: database triggers keep it balanced and immutable

Accounts
bcrypt, short-lived signed sessions, instant revocation, lockout without account enumeration.

Audit
Every agent step and every business action recorded: who, what, when, before and after.

Compliance
SYSCOHADA revised chart, VAT returns, closed periods, segregation of duties between cash handlers and owner.

PROOF THAT IT WORKS

Tested like a bank would expect

15 / 15

agent scenarios with the real model on a 28-week sandbox business; 145 audit events

1.3 s

median model step; full answer in about 5 s

443

unit and integration tests on a real database, plus 76 browser tests (13 end-to-end suites, offline)

400

simulated users in 60 businesses: 3 738 requests, 0 errors

What the sandbox found and we fixed

The AI provider retired the default model, so we switched models and made the model a setting. Overloaded AI responses are now retried. A cash request rule the agent did not know is now applied before any proposal.

What we disclose

Two answers came in French to English questions; one figure was multiplied by the model (correctly). Both are on the known-risk list with a fix planned.

TECHNOLOGY

Engineered for weak networks and many businesses at once

Offline-first

Pages and code saved per session; an outbox in the browser; sync in order with retries. A sale made without internet is recorded exactly once when the connection returns.

Multi-tenant

Next.js 16 and React 19 on Vercel, PostgreSQL on Supabase, 101 data models, 13 department types, background jobs for reports and the ledger.

Model-agnostic agent

Google Gemini with function calling behind a small adapter, so the model is a setting. Tools are plain server functions with their own guards. Fully audited.

About 57 000 lines of application code and 11 000 lines of tests, written and maintained by team Neur Artisa.

Next.js 16 · React 19 | **PostgreSQL · Prisma** | **Vercel · Supabase · Inngest** | **Gemini function calling**

BUSINESS MODEL

A monthly subscription per business

SUBSCRIPTION

20 000 FCFA

per business per month, as the entry price: every department, the Boss's view, offline use, reports and the copilot.

Illustration: **1 000 businesses = 240 million FCFA a year** of recurring revenue.

Go to market

Pilot with 3–5 businesses in Yaoundé, then word of mouth among owners, accountants who serve several SMEs, and associations of traders.

Expansion revenue

Full accounting and VAT for growing businesses; mobile-money collection and reconciliation; financing referrals once lenders accept the records.

Why they pay

One recovered cash shortfall or one collected old debt a month covers the subscription.

WHY SPRINGER FINANCE

Made for how these businesses actually work

	Notebook / spreadsheet	Generic accounting software	Springer Finance
Several activities, each with its own screens	—	partly	13 types
Cash each manager should hold, live	—	—	yes
Works without internet, syncs later	yes	rarely	yes
No accountant needed; SYSCOHADA when ready	no rules	accountant needed	both
FCFA, mobile money, credit sales, tontines	by hand	partly	built in
AI copilot that analyses and prepares records	—	—	with approval

ROADMAP

From working product to paying customers

NEXT MONTH

Ready for pilots

- Paid AI key (no training use of data)
- CI, error tracking, backups
- First 3–5 pilot businesses

QUARTER 1

Local fit

- Full French interface
- MTN MoMo and Orange Money APIs
- Debt reminders by SMS / WhatsApp, proposed by the copilot

QUARTER 2

Mobile and paid plans

- Installable app, then Play Store
- Subscriptions live
- More copilot tools (supplier bills)

QUARTER 3+

Financing bridge

- Records accepted by partner microfinance institutions
- CEMAC and wider francophone Africa

Today: 13 department types live · copilot with 11 tools · full test suites green

TEAM AND ASK

Team Neur Artisa: built in Cameroon, for Cameroonian businesses first

Alloh Nkefor Rodrick · team leader

Master's student in Computer Engineering, National Higher Polytechnic Institute (NAHPI), University of Bamenda. Full-stack and AI engineer; designed and built Springer Finance.

Alloh Springere Ngonjoh

Second-year Biomedical Engineering student, National Higher Polytechnic Institute (NAHPI), University of Bamenda.

What we are looking for

Feedback from WeBank's experts on the agent's design; introductions to microfinance partners and pilot businesses; cloud and AI credits to run the copilot for pilots.

Thank you. **Springer Finance**: every franc of the business, understood.

Repository: github.com/NobleRodrick/nonchalant-finance